

Venoarterial Extracorporeal Membrane Oxygenation Versus Conventional Therapy in Severe Pediatric Septic Shock*

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Objectives: The role of venoarterial extracorporeal membrane oxygenation in the treatment of severe pediatric septic shock continues to be intensely debated. Our objective was to determine whether the use of venoarterial extracorporeal membrane oxygenation in severe septic shock was associated with altered patient mortality, morbidity, and/or length of ICU and hospital stay when compared with conventional therapy.

Design: International multicenter, retrospective cohort study using prospectively collected data of children admitted to intensive care with a diagnosis of severe septic shock between the years 2006 and 2014.

Setting: Tertiary PICUs in Australia, New Zealand, Netherlands, United Kingdom, and United States.

Patients: Children greater than 30 days old and less than 18 years old.

Interventions: None.

Measurements and Main Results: Of 2,452 children with a diagnosis of sepsis or septic shock, 164 patients met the inclusion criteria for severe septic shock. With conventional therapy ($n = 120$), survival to hospital discharge was 40%. With venoarterial extracorporeal membrane oxygenation ($n = 44$), survival was 50% ($p = 0.25$; CI, -0.3 to 0.1). In children who suffered an in-hospital cardiac arrest, survival to hospital discharge was 18% with conventional therapy and 42% with venoarterial extracorporeal membrane oxygenation ($\Delta = 24\%$; $p = 0.02$; CI, 2.5 – 42%). Survival was significantly higher in patients who received high extracorporeal membrane oxygenation flows of greater than 150 mL/kg/min compared with children who received standard extracorporeal membrane oxygenation flows or no extracorporeal membrane oxygenation (82%, 43%, and 48%; $p = 0.03$; CI, 0.1 – 0.7 and $p < 0.01$; CI, 0.2 – 0.7 , respectively). Lengths of ICU and hospital stay were significantly longer for children who had venoarterial extracorporeal membrane oxygenation.

Conclusions: The use of venoarterial extracorporeal membrane oxygenation in severe pediatric sepsis is not by itself associated with improved survival. However, venoarterial extracorporeal membrane oxygenation significantly reduces mortality after cardiac arrest due to septic shock. Venoarterial extracorporeal membrane oxygenation flows greater than 150 mL/kg/min are associated with almost twice the survival rate of conventional therapy or standard-flow extracorporeal membrane oxygenation. (*Pediatr Crit Care Med* 2018; 19:965–972)

Key Words: acute septic shock; cardiac arrest; extracorporeal membrane oxygenation; septic shock; survival rate; treatment

Septic shock is defined as sepsis with resulting organ dysfunction (1, 2). Severe sepsis and shock remain a significant causes of morbidity and mortality in pediatric intensive care despite decades of gradual advances in therapy and well-established management guidelines (2–4). The occurrence rate of severe sepsis appears to be increasing (5, 6). In addition to the timely administration of antibiotics, the early and vigorous pursuit of circulatory stabilization in septic shock continue to be the focus of efforts to improve patient survival (5, 7, 8). At the same time, there remains significant variability in the management of sepsis between ICUs across the world (9). Although a number of these differences may be attributed to varying levels of guideline implementation or local resources, challenges to the validity of the evidence behind some recommendations and interventions feature prominently (10, 11).

The role of venoarterial extracorporeal membrane oxygenation (VA ECMO) in the treatment of severe pediatric sepsis and shock is an important example of an intervention that continues to be intensely debated. Reports of the use of VA ECMO in children with severe sepsis in the 1990s and early 2000s described the safety of ECMO in this clinical scenario and later established that ECMO should no longer be seen as contraindicated in septic shock (12–16). More recent reports from single centers contend that VA ECMO may provide better survival than conventional therapy and furthermore that strategies to maximize flow rates in order to reverse shock and multiple organ dysfunction may play an important role (17, 18). Recommendations regarding blood flow during VA ECMO in septic shock, however, are also subject to controversy with some authors cautioning against delivering more than 110 mL/kg/min, whereas others recommend more than 150 mL/kg/min (3).

However, whether the use of ECMO in septic shock confers a true outcome benefit has been questioned early on (19, 20) and continues to remain in doubt (21). In addition, there is considerable concern about the risks of this highly invasive therapy such as major hemorrhage and thromboembolic events. Consequently, the most recent international guideline for the management of severe sepsis and septic shock in children specifies ECMO as a therapy of last resort for refractory shock with a grading of 2C, designating it a weak recommendation with a low level of evidence (2).

In this context, the aim of our study was to examine whether the use of VA ECMO in severe pediatric septic shock was associated with patient outcome.

MATERIALS AND METHODS

We conducted an international multicenter, retrospective cohort study of children admitted to intensive care between the years 2006 and 2014, seeking to determine whether the use of mechanical circulatory support in severe septic shock was associated with altered patient mortality, morbidity, and/or length of ICU and hospital stay when compared with conventional therapy. The participating seven PICUs (A–G) identified children with a diagnosis of sepsis or septic shock in their respective patient databases. Data in each unit patient database had been contemporaneously collected and stored during the patients' stay in intensive care. Patients were included in the study if they were greater than 30 days old and less than 18 years old, had a positive microbial culture or had tested positive for a pathogen on polymerase chain reaction, and were intubated as well as mechanically ventilated and received inotropes. In addition, inclusion in the study required children to meet three out of four criteria for severe septic shock in the first 24 hours of their ICU stay: arterial pH less than or equal to 7.15, arterial lactate greater than or equal to 4.0, base excess less than or equal to –10, and in-hospital cardiac arrest. Patients were excluded if they had cyanotic congenital heart disease, myocarditis, or an out-of-hospital cardiac arrest.

Participating centers were required to have an established ECMO program. The selection of centers aimed to include programs with different approaches to the use of extracorporeal support in severe septic shock, ranging from routine use to occasional application to not offering ECMO for this indication. Specific criteria for the use of ECMO in severe septic shock, however, were not in existence. In order to join the study, each center was required to identify and submit complete data for at least 10 eligible patients. Each unit obtained approval for conducting the study from their respective research ethics board.

In addition to deidentified demographic data and data relating to inclusion and exclusion criteria, information collected comprised the presence of any preexisting condition, ventilator and hemodynamic indicators as well as ECMO parameters at 4 hours after institution of extracorporeal circulatory support, and the time from admission to ICU to institution of ECMO. Lactate, base excess, and pH were recorded as highest and lowest, respectively, in the first 24 hours of ICU stay. The primary outcome measure was survival to hospital discharge. Secondary outcomes were time to resolution of major organ failure (length of respiratory and inotropic support), the need for renal replacement therapy, ICU and hospital length of stay, a need for a long-term tracheostomy, the requirement of long-term hemodialysis, and permanent ischemic limb injury. The minimum $\text{PaO}_2/\text{FiO}_2$ ratio, minimum mean arterial pressure, and maximum inotrope doses during the ICU stay were ascertained. The Vasoactive-Inotrope Score was calculated as described by Gaies et al (22).

Patients were cohorted by conventional therapy versus VA ECMO. Subsequently, the VA ECMO cohort was divided by level of mechanical support into patients who received high-flow VA ECMO (≥ 150 mL/kg/min at 4 hr of therapy) and patients who received standard-flow VA ECMO (< 150 mL/kg/min at 4 hr of therapy). For children who had a cardiac arrest and received

VA ECMO, the ECMO classification that participating units had reported to the registry of the Extracorporeal Life Support Organization (ELSO) was determined in order to ascertain whether these patients had been categorized as extracorporeal cardiopulmonary resuscitation (ECPR). Quantile regression analysis was performed to compare medians across two groups, and Kruskal Wallis tests were performed to compare three groups since the distribution of continuous variables was skewed. The Chi-square test was used to compare proportions of two and three groups. Logistic regression was used in the comparison of four groups and to estimate predictors of death across all groups, adjusted for unit (PICUs A to G) in order to control for clustering. In case of missing data, patients were excluded from analysis. The statistical approach was reviewed and statistical analysis

performed by the Clinical Epidemiology and Biostatistics Unit at the Murdoch Children's Research Institute (Melbourne, Australia). Calculations were performed with a commercial software package (Stata 14, StataCorp LP, College Station, TX).

RESULTS

Patient Cohorts

The seven participating PICUs identified 2,452 children with a diagnosis of sepsis or septic shock from January 1, 2006, to December 31, 2014. Of these children, 936 (38%) received invasive mechanical ventilation as well as inotropes. One hundred sixty-four patients (6.7%) subsequently met at least three out of the four criteria for severe septic shock and were eligible for inclusion in the study (Fig. 1).

Of the patients included in the study, 44 (27%) had received VA ECMO, whereas 120 (73%) had conventional therapy without extracorporeal circulatory support. Patient characteristics were similar in the two patient cohorts, including the level of physiologic derangement reflected in the criteria for severe septic shock (Table 1). The proportion of children with and without life-limiting conditions was likewise similar in the two groups. However, there were only two patients with a malignancy/immunosuppressive disease in the VA ECMO cohort versus 26 in the group that had received conventional therapy.

Pathogens

Infections with a single bacterial pathogen dominated (73%) while mixed-growth, fungal, and viral infections were found in the remaining 27% of patients (Fig. S1, Supplemental Digital Content 1, <http://links.lww.com/PCC/A698>; legend, Supplemental Digital Content 5, <http://links.lww.com/PCC/A702>). Staphylococci were responsible for the majority of cases of Gram-positive sepsis (*Staphylococcus aureus* 44%, coagulase-negative *Staphylococcus* 7%).

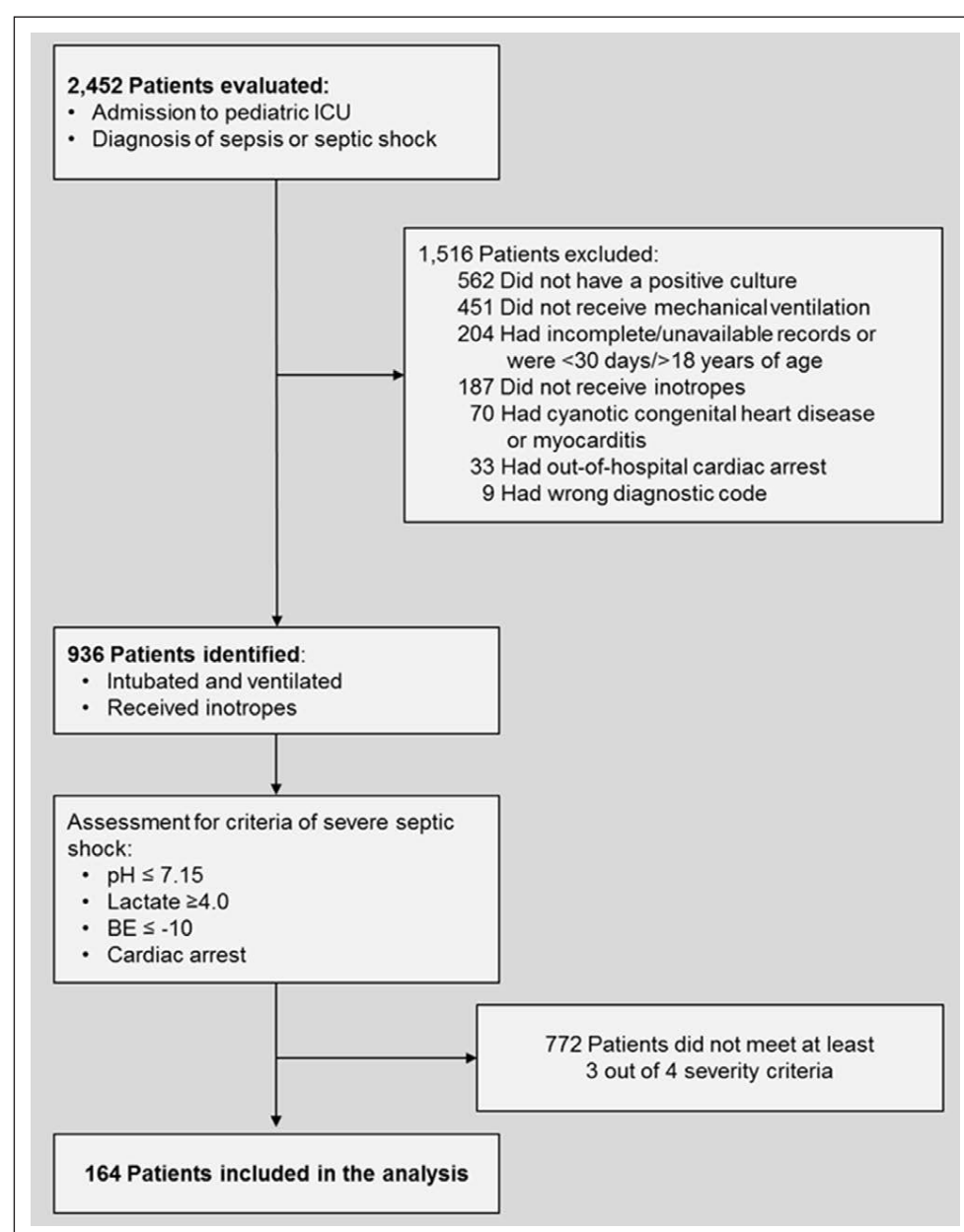


Figure 1. Patient enrolment and assessment. BE = base excess.

TABLE 1. Baseline Patient Characteristics

Variable	Patients With Conventional Therapy (n = 120)	Patients With Venoarterial Extracorporeal Membrane Oxygenation Therapy (n = 44)	Comparison, 95% CI (p)
Age, yr, median (IQR)	5.1 (1.5–12.2)	3.6 (0.5–8.0)	–5.1 to 2.3 (0.47)
Weight, kg, median (IQR)	16.9 (10.7–40.9)	19.0 (0.5–29.9)	–5.9 to 11.9 (0.51)
Male sex, n (%)	59 (49)	23 (52)	–0.1 to 0.2 (0.72)
pH, median (IQR)	7.04 (6.9–7.1)	7.01 (6.9–7.1)	–0.1 to 0.0 (0.27)
Lactate, arterial (mmol/L), median (IQR) ^a	8.7 (5.3–12.7)	9.1 (6.1–12.5)	–1.9 to 2.7 (0.73)
Base excess, median (IQR)	–16.0 (–20.0 to –12.3)	–15.0 (–19.0 to –11.0)	–1.2 to 3.2 (0.36)
Cardiac arrest, n (%)	56 (47)	26 (59)	–0.3 to 0.1 (0.16)
No preexisting, life-limiting condition, n (%)	62 (52)	28 (64)	–0.3 to 0.1 (0.17)
Malignancy/immunosuppression	26 (22)	2 (5)	7.8 to 26.7 (0.01)
Other preexisting, life-limiting condition, number (%)	32 (27)	14 (32)	–0.2 to 0.1 (0.52)

IQR = interquartile range.

^an = 114.

Streptococci caused 49% of cases of Gram-positive sepsis. Among the cases of Gram-negative sepsis, 68% were caused by coliforms and 20% by *Pseudomonas* species, whereas *Stenotrophomonas*, *Haemophilus influenzae*, and one instance of *Shigella* had been identified in the remainder of cases.

Viral sepsis had been caused by influenza A and *Enterovirus* in four cases each. In the other four cases of viral sepsis, *Cytomegalovirus* (two cases), *Human Metapneumovirus* (one case), and *Parainfluenza* (one case) had been found. *Candida* species were responsible for all instances of fungal sepsis as well as mixed fungal-bacterial infections (eight cases of *Candida albicans* and one case each of *Candida parapsilosis*, *Candida lusitanae*, and *Candida tropicalis*).

Primary Outcome

Patient survival to hospital discharge varied markedly between participating centers and did not display any obvious correlation with the frequency of ECMO use (Fig. 2). There was no significant difference in survival to hospital discharge between patients who had received VA ECMO and those who had had conventional therapy. In the cohort that had conventional therapy, 48 patients (40%) survived to hospital discharge while there were 22 survivors (50%) in the group that had received VA ECMO (95% CI, –30 to 10%; $p = 0.25$).

Across all patients, the outcome was worst for children with a malignancy/immunosuppression ($n = 28$) compared with patients with other preexisting conditions ($n = 46$) and previously well children ($n = 90$) with survival rates to hospital discharge of 21%, 43%, and 49%, respectively ($p = 0.04$).

There was a significant difference between cohorts of children who had suffered an in-hospital cardiac arrest: survival to hospital discharge was 18% with conventional therapy and 42% with VA ECMO ($\Delta = 24\%$; 95% CI, 2.5–42%; $p = 0.02$). When children with a malignancy/immunosuppression were

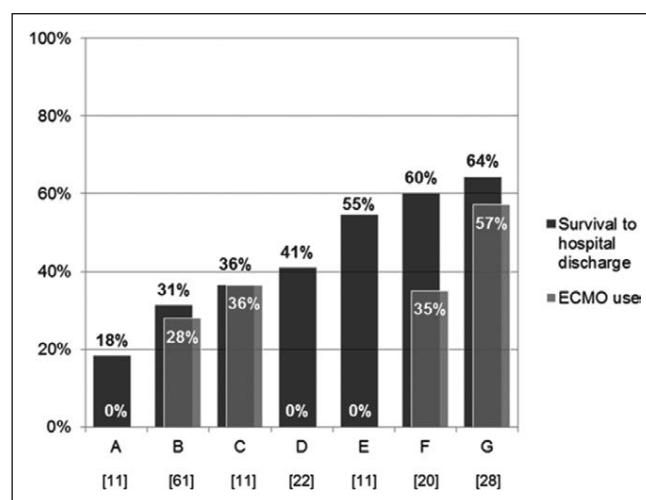


Figure 2. Patient survival to hospital discharge (dark gray) and proportion of venoarterial extracorporeal membrane oxygenation (ECMO) use (light gray) in seven PICUs (A–G). Numbers in brackets = number of patients enrolled by unit.

excluded from the analysis, the difference in survival between the cohorts reduced to 22% (95% CI, –50 to 0%; $p = 0.05$) (Fig. 3). Children, who in the course of their illness suffered an in-hospital cardiac arrest and were placed on extracorporeal life support, had been in the ICU for a median time of 5.0 hours before receiving ECMO (interquartile range [IQR], 1.7–12.8). Of these children, 18 (69%) had been reported to the ELSO registry as having received ECPR, whereas five patients (19%) received the ECMO category “cardiac” and three patients (12%) the category “pulmonary.”

Analyzing patient primary outcome by level of mechanical circulatory support, survival was significantly higher in patients who received high ECMO flows (> 150 mL/kg/min at 4 hr after institution of extracorporeal support) compared with

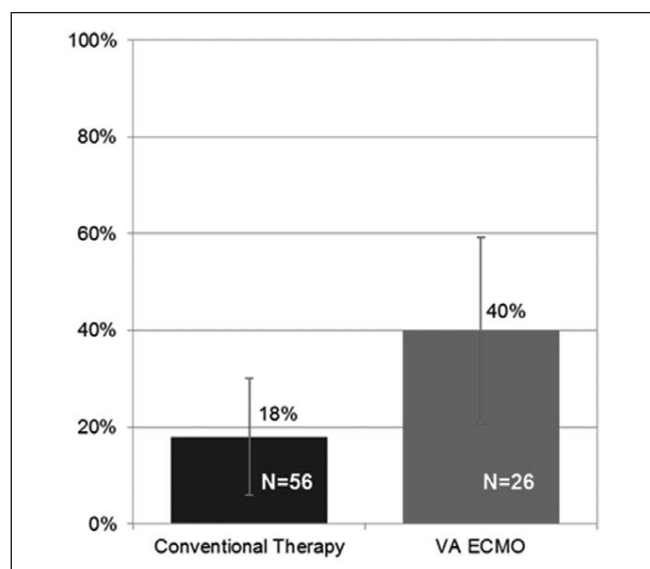


Figure 3. Patient survival after cardiac arrest (excluding children with malignancy/immunosuppression). VA ECMO = venoarterial extracorporeal membrane oxygenation.

children who received standard ECMO flows or no ECMO ($\Delta = 39\%$; 95% CI, 0.1–0.7; $p = 0.03$ and $\Delta = 40\%$; 95% CI, 0.2–0.7; $p < 0.01$, respectively). Patient demographics, physiologic derangement (pH, lactate, base excess), occurrence rate of cardiac arrest, Vasoactive-Inotropic Scores, and mean airway pressures were similar across the three groups (Table S1, Supplemental Digital Content 2, <http://links.lww.com/PCC/A699>). When examined at 24 hours postinitiation of ECMO, high flows greater than 150 mL/kg/min had been maintained in nine of 11 patients (82%). In two patients, ECMO flow had declined from 178 to 144 mL/kg/min and from 175 to 124 mL/kg/min, respectively. Both these patients survived. There was no evidence to suggest a difference in survival between children on standard-flow VA ECMO (< 150 mL/kg/min at 4 hr) and children who did not receive ECMO ($\Delta = 3\%$; 95% CI, 0–20%; $p = 0.74$). These results remained unchanged when patients with a malignancy/immunosuppression were excluded (Fig. 4). Similarly, the odds of death were higher in children with ECMO flows up to 110 mL/kg/min as well as between 110 and 150 mL/kg/min compared with ECMO flow of over 150 mL/kg/min (Table 2). The nine survivors in the high-flow ECMO cohort had central (atrio-aortic) cannulation through the open chest, whereas the two nonsurvivors had been cannulated peripherally (jugulocrotid). Central cannulation was less frequent in the standard-flow ECMO cohort ($\Delta = 39\%$; 95% CI, 16–67%; $p = 0.003$).

Secondary Outcomes

The cohort of children who received VA ECMO had a significantly longer duration of invasive respiratory as well as inotropic support. The median (IQR) time from admission to ICU to institution of VA ECMO was 7.8 (2.5–24) hours. This group also had longer lengths of stay in ICU as well as in hospital and a higher proportion of patients receiving renal replacement therapy. The two cohorts showed no evidence of difference in

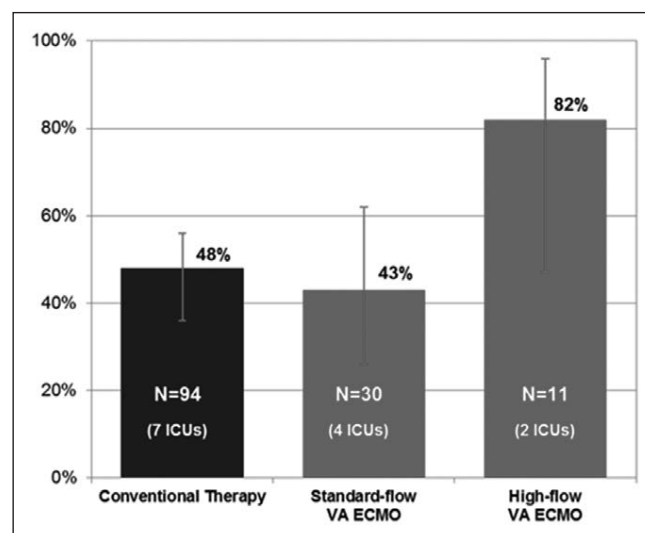


Figure 4. Patient survival by level of mechanical circulatory support (excluding children with malignancy/immunosuppression): conventional therapy, standard-flow venoarterial extracorporeal membrane oxygenation (VA ECMO) (< 150 mL/kg/min at 4 hr), and high-flow VA ECMO (≥ 150 mL/kg/min at 4 hr).

TABLE 2. Odds of Death by Level of Mechanical Circulatory Support

Extracorporeal Membrane Oxygenation Flow at 4 hr of Therapy in mL/kg/min	n	OR ^a (95% CI)	p
≥ 150	11	1	—
110 to < 150	15	6.4 (0.9–42.8)	0.06
< 110	15	4.4 (0.6–32.4)	0.14
None (conventional therapy)	120	5.0 (0.9–27.2)	0.06

OR = odds ratio.

^aAdjusted for unit (PICUs A to G).

P_{aO_2}/F_{iO_2} ratios, mean airway pressures, Vasoactive-Inotrope Scores, or mean arterial pressures (Table S2, Supplemental Digital Content 3, <http://links.lww.com/PCC/A700>).

In survivors, the differences between the two cohorts in duration of ventilatory support, length of ICU stay, or length of hospital stay were not significant (Table S3, Supplemental Digital Content 4, <http://links.lww.com/PCC/A701>). Tracheostomies were rare events (one tracheostomy in the cohort with conventional therapy, two tracheostomies in patients in the VA ECMO cohort). Similarly, permanent ischemic limb injury was uncommon (five occurrences with conventional therapy, two with VA ECMO). Renal replacement therapy was used significantly more often in the ECMO cohort.

Predictors of Poor Outcome

The variables for the multivariate logistics regression analysis to predict status at discharge were pH, lactate, and base excess. Occurrence of cardiac arrest, malignancy/immunosuppression, and unit (PICUs A–G) were considered as covariates. The level of lactate and occurrence of in-hospital cardiac arrest

were predictive of death (odds ratio [OR], 1.2; 95% CI, 1.1–1.4 and OR, 4.3; 95% CI, 1.9–9.7, respectively). Neither low pH (< 7.05) nor base excess correlated with death (OR, 1.3; 95% CI, 0.6–3.2 and OR, 1.0; 95% CI, 0.9–1.1, respectively). Similarly, a malignancy/immunosuppression was not in itself a predictor of death (OR, 2.4; 95% CI, 0.7–8.4), nor was the presence of other preexisting conditions (OR, 2.3; 95% CI, 0.9–5.9)

DISCUSSION

Management of circulatory failure is the centerpiece of intensive care therapy in severe septic shock (23). Employing extracorporeal support to augment or substitute native cardiac output in this setting is therefore a logical strategy. However, whether the theoretical benefits of VA ECMO outweigh the hazards of this highly invasive therapy and, importantly, result in a clinical advantage over modern conventional ICU therapy remains to be determined.

The characteristics of the patient population and the epidemiology of pediatric sepsis in our large, multicenter study were similar to what has previously been reported in the literature (6, 24). Consistent with established mortality prediction models and PICU outcome research, we found that serum lactate levels and cardiac arrest correlated with death (25, 26). Against this background, our analysis showed that the frequency of VA ECMO use in severe pediatric sepsis was not by itself associated with improved outcome: patient survival did not correlate with how commonly a PICU employed VA ECMO in severe septic shock. In apparent keeping with this finding was the absence of a significant difference in mortality and morbidity between the cohort of children who received VA ECMO and the cohort who had conventional therapy.

VA ECMO was associated with increased length of ICU and hospital stay. This association was not present in survivors, indicating that VA ECMO resulted in nonsurvivors being kept alive for longer. Based on these results, VA ECMO did not appear to offer an advantage over conventional therapy but increased ICU and hospital resource use. However, this negative finding should be treated with some caution. Despite failing to reach statistical significance, the absolute difference in mortality of 10% between the two cohorts may be deemed clinically important, particularly in light of our study's limits with regard to sample size and the relatively wide CI.

Extracorporeal support, on the other hand, is commonly regarded a rescue therapy (2) with a consequent selection bias for more severely ill patients. The fact that survival in the VA ECMO cohort was similar to conventional therapy may therefore in effect indicate a therapeutic gain. Such a selection bias, however, was not obvious in the analysis of the objective measures of severity of illness and intensity of conventional organ support in the two cohorts. There was no evidence of significant difference in physiologic derangement (pH, lactate, base excess), in the occurrence rate of cardiac arrest, in Vasoactive-Inotrope Scores, or mean airway pressures. Whether these data truly encompasses the severity and complexity of a bedside "rescue situation" and therefore rules out any selection bias is,

nevertheless, debatable. Our results therefore remain inconclusive on this point.

In contrast, the effect of VA ECMO was marked in children who had suffered an in-hospital cardiac arrest. In this scenario, extracorporeal cardiorespiratory support was associated with a 24% survival advantage. Although this benefit decreased and reached the limit of statistical significance when negative selection bias for patients with a malignancy/immunosuppression was excluded, it remained substantial at 22%. This result is in keeping with mounting evidence about the effectiveness of extracorporeal circulatory support in cardiac arrest in a variety of scenarios in both children and adults. In a recent meta-analysis, Kim et al (27) found superior survival and neurologic outcome at 3–6 months in adults after ECPR compared with conventional resuscitation after in-hospital cardiac arrest. This survival advantage was independent of the etiology of cardiac arrest. Earlier this decade in a large registry study, Raymond et al (28) had reported a 44% survival rate with VA ECMO in children who had failed conventional resuscitation. The majority of survivors, however, had a primary cardiac pathology and outcome in children with septic shock had been poor. Our large cohort study shows that a survival rate of 40% can be achieved with VA ECMO after cardiac arrest in profound pediatric septic shock. It should be noted, however, that this includes a sizable minority of children who had not received an ELSO registry classification of ECPR and who therefore may be presumed to have had return of spontaneous circulation at the time of initiation of ECMO.

The most substantial reduction in mortality from severe pediatric septic shock, however, was seen with VA ECMO flows greater than 150 mL/kg/min at 4 hours after implementation of extracorporeal support. The survival rate of children who received these very high-flow rates in two of the seven participating PICUs was almost double that seen in all other patients in our study. This finding is consistent with previous reports from single centers (17, 18) and supports the proposition that maximizing extracorporeal circulatory output is fundamental to survival. The need for high ECMO flow may also explain why standard-flow VA ECMO did not yield higher survival rates when compared with conventional therapy. Contrary to concerns that VA ECMO blood flow above 110 mL/kg/hr may be detrimental (3), our data show that the odds of death are in fact higher below this threshold. The wide CIs around this measurement, however, indicate that its precision is not high.

Our study has both limitations and strengths. Intensive care therapy in profound septic shock is fast-paced and complex. In this context, there remains a significant potential for a retrospective analysis to be subject to a range of confounders. We aimed to reduce these through strict definitions and inclusion criteria. Although this approach reduces unmeasured variables arising across different practice settings, it cannot eliminate them. In addition, factors extraneous to the PICU and individual patients such as burden of disease in the community or access and presentation to tertiary healthcare may have influenced outcomes.

The single primary outcome measure was survival, and *p* values were not adjusted for multiple secondary outcomes. This approach was chosen to balance the risk of type 2 errors against the probability of type 1 errors in light of the magnitude of effects (29). With respect to pharmacologic treatment of shock, we calculated the Vasoactive-Inotropic Score (22). Although this is an established approach that yields an objective aggregate comparator, it does not allow distinguishing between vasoplegic and vasoconstrictive shock states that may potentially have different responses to mechanical circulatory support.

Our analysis represents the largest cohort study comparing extracorporeal circulatory support and conventional therapy in severe pediatric sepsis. We sought to elucidate the influence of one particular treatment, VA ECMO, on the outcome of septic shock, a multifaceted clinical syndrome, across a number of diverse PICUs in five countries. This approach required any treatment effect to be large in order to stand out against the influences of a broad range of therapies and interventions delivered to children with severe sepsis. This also means that our results can only be interpreted as embedded in the overall practice patterns of the participating PICUs. The reduced mortality with VA ECMO after cardiac arrest and the impressive survival benefit seen with high-flow ECMO in septic shock may be due not to this single therapeutic modality but instead to systems of intensive care that combine the range of available therapies into the most effective overall treatment strategy. For example, our data cannot rule out that the significantly more frequent use of renal replacement therapy in the ECMO cohort contributed to a difference in outcome. Similarly, it should be noted that high-flow ECMO was delivered predominantly by central, atrio-aortic cannulation. This technique poses distinct risks and challenges. These include acute and subacute thoracic hemorrhage, a frequent need for surgical reexploration in the first 24–36 hours and the risk of mediastinitis. It reflects a particular approach to extracorporeal support that would not easily be replicated in every ECMO program. At the same time, it may also in itself have an impact on morbidity as well as PICU and hospital length of stay. Our study therefore represents a comparison of clinical effectiveness of different approaches to therapy. It was not designed to determine long-term functional patient outcomes or to provide a cost-effectiveness analysis.

CONCLUSIONS

The use of extracorporeal circulatory support in severe pediatric sepsis is not by itself associated with improved survival. However, VA ECMO significantly reduces mortality after cardiac arrest due to septic shock. VA ECMO with flows of 150 mL/kg/min or greater is associated with almost twice the survival rate of conventional therapy or standard-flow ECMO.

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